

SMART XEROX MANAGEMENT SYSTEM

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Abstract— Manual order processing and physical lines are the mainstays of traditional Xerox services, which result in inefficiencies and privacy issues. The Smart On-Demand Xerox Management System described in this paper uses a safe web-based platform to digitize the printing process. Uploading documents, choosing print options, paying online, and monitoring order status in real time are all available to users. The Student, Host, and Admin modules are part of a role-based architecture. While automated file deletion improves privacy, OTP authentication guarantees security. The system, which was created with the help of Node.js, Express.js, MongoDB, and Razorpay integration, increases document printing services' scalability, efficiency, and transparency.

Keywords— Role-Based Access Control, Digital Printing Platform, Web-Based Application, Online Payment Integration, OTP Authentication, Document Privacy, Real-Time Order Tracking, FastTrack Priority Scheduling, MongoDB Database, Smart Xerox Management System.

1. INTRODUCTION

Digital technologies are always changing, and this has had a big impact on service-based industries by making them more efficient, easier to reach, and more open about how they work. Even though technology is moving quickly, most of the time, traditional document printing and Xerox services still rely on manual processes. In a lot of schools, offices, and businesses, people have to go to printing shops in person, wait in line, tell the staff what they need to print, and pay with cash. Using these traditional methods can lead to long wait times, confusion about orders during busy times, limited tracking options, and a lack of systematic record keeping. Also, sensitive documents like academic records, official forms, and private reports could still be stored or accessed without structured data handling policies, which could put privacy at risk.

As more and more people want contactless services and digital convenience, it's important to update the old printing process into a more organized and tech-driven system. Combining web-based platforms, secure authentication methods, and online payment gateways is a useful way to fix problems with manual printing services. Digital platforms can greatly cut down on physical crowding, mistakes made by people, and overall service reliability by letting people

submit documents from afar, process orders automatically, and track orders in real time. Also, adding secure user authentication and controlled document lifecycle management helps protect privacy and keep transactions honest.

From the perspective of technology, the development of the platform also proves the effective use of modern web development tools and technology, database management tools, and authentication technology. The use of these tools and technology together allows the system to perform the task of user authentication, upload of files, verification of payments, and processing of orders in an efficient way.

Thus, the Smart On-Demand Xerox Management System is an example of the effective use of digital transformation technology in the management of day-to-day activities of service industries. The challenges faced by the users of the printing shops and the owners of the shops have been addressed in an effective way. The proposed method will not only be effective for the users and the owners of the shops but will also lay the groundwork for the expansion of the system to cover the entire distributed printing industry.

This paper proposes the design and implementation of a Smart On-Demand Xerox Management System that turns traditional printing processes into a structured digital ecosystem in response to these problems. The system is built around roles, with modules for students, hosts, and admins. This makes it easy to place, process, and keep track of orders. Users can upload files from anywhere, choose how they want to print, make safe online payments, and see the status of their orders in real time. The implementation includes OTP-based authentication to verify users, automated file deletion to improve data security, and analytical dashboards to keep an eye on revenue and orders. The proposed solution aims to improve operational efficiency, ensure transparency, and provide a scalable framework that can support multi-location printing services in future deployments by combining modern web technologies with secure database management.

2. LITERATURE SURVEY

Recent research in digital service automation has greatly increased the efficiency of small businesses by bringing together web-based platforms, secure authentication systems, and online payment gateways. A number of studies have looked into queue management systems, role-based web apps, and digital transaction frameworks from 2021 to 2025 to cut down on manual work and make services more open. Nonetheless, there exists a paucity of research concentrated on the modernization of conventional document printing and Xerox services. This section looks at fifteen studies that are relevant and compares their methods to the Smart On-Demand Xerox Management System that is being proposed.

Ramesh et al. [1] put forward a web-based queue management system that uses FIFO scheduling. This cut down on the time customers had to wait in public service centers by 32%. Their research showed that structured queue processing greatly increases the number of services that can be provided. Iyer and Sharma [2] also made a digital service management platform with role-based access control. They reported that 91% of users were happy with it because it made things clearer and more organized. Their results show how important authentication and access segmentation are. Our system uses OTP-based login for both Students and Hosts to do this. Mehta et al. [3] presented an online order processing framework incorporating payment verification APIs, achieving a 98% success rate in transaction validation. Their focus on payment-first confirmation fits with how our system works: the host can only access documents after Razorpay verification has been successful. Kumar and Nair [4] made a retail monitoring system based on a dashboard. They saw a 27% increase in operational efficiency thanks to real-time analytics and revenue visualization. Our Admin panel combines revenue metrics, order distribution statistics, and FastTrack usage monitoring, which is similar to this method. Khan et al. [5] looked into how to handle documents and found that cloud-based document submission platforms had privacy risks because files were stored for a long time. They suggested automating file deletion to prevent misuse. To get around this problem, our system deletes PDFs after a certain amount of time has passed since the order was placed. Patel et al. [6] examined scalable web architectures utilizing Node.js and MongoDB, illustrating the capacity to accommodate concurrent users with response times 35% faster than conventional server frameworks. This confirms that we made the right choice by using Node.js and MongoDB to process orders in real time.

Singh et al. [7] compared manual service workflows to digital platforms and found that automation cut down on processing errors by 40%. Hussein et al. [8] created a web-based service application that could handle priorities, which made it 29% faster to finish urgent tasks. This backs up our plan to use FastTrack priority scheduling with FIFO order handling. Rao et al. [9] created a multi-role service management application that showed better system accountability and a 22% rise in order accuracy. Awad and Verma [10] looked at secure authentication methods in web apps and showed that using one-time passwords (OTPs) for verification cuts down on unauthorized access incidents by 31%. This backs up the security model that our system uses. Johnson et al. [11] did a meta-analysis of digital transaction systems and found that payment gateways that work with APIs have reliability rates above 97%. This proves that Razorpay is a safe way to make online payments. Swathy et al. [12] investigated analytics-driven dashboards for small businesses and found that they made it 34% easier to keep track of finances. Ogunpola et al. [13] stressed the need for modular web architecture to make systems more scalable, which led to a 30% increase in adaptability during upgrades. Al'Aref et al. [14] examined user-centric digital platforms, emphasizing personalization and transparency as essential factors for adoption. Lastly, Johnson and Lee [15] looked at automation in small business services and found that integrated digital platforms make operations 25%–45% more efficient than manual systems.

Most of the fifteen studies reviewed focus on one of the following topics: queue optimization, payment integration, authentication security, or analytics dashboards. However, none of the systems that were looked at put all of the parts—secure OTP authentication, payment-first validation, FIFO and FastTrack scheduling, role-based dashboards, real-time analytics, and automated document deletion—into a single, unified framework that is made for document printing services. The suggested Smart On-Demand Xerox Management System fills this gap by putting all of these features into a digital ecosystem that can grow and is safe.

The literature generally agrees that web-based automation makes things much more efficient, clear, and easy for users. The proposed system builds on these well-known ideas by adding document lifecycle management and privacy enforcement. This is a new area of research in service digitization that hasn't been looked into much yet.

3. PROPOSED SYSTEM

The proposed Smart On-Demand Xerox Management System is intended to digitalize and automate conventional printing services through a secure, scalable, and role-based web architecture. The system can minimize the need for human operators to manage queues, reduce waiting time, and ensure the security of document processing through authentication and automated workflow systems. The system is built with Node.js and Express.js technologies for backend services, MongoDB for database management, and a web-based frontend interface for user interaction.

3.1 System Architecture

The Smart On-Demand Xerox Management System architecture follows a client-server architecture, where the frontend, implemented with HTML, CSS, and JavaScript, communicates with the backend, implemented with Node.js and Express.js. The system architecture is divided into three layers to ensure an organized system development and efficient system performance. The Presentation Layer deals with the implementation of role-based dashboards for Students, Hosts, and Admin users. The Application Layer handles authentication, order processing, payment verification, and priority scheduling through FIFO and FastTrack. The Data Layer manages user accounts, order processing, payment, and document metadata through MongoDB.

ensemble of decision trees to obtain high accuracy and efficiency in prediction. XGBoost optimizes gradient

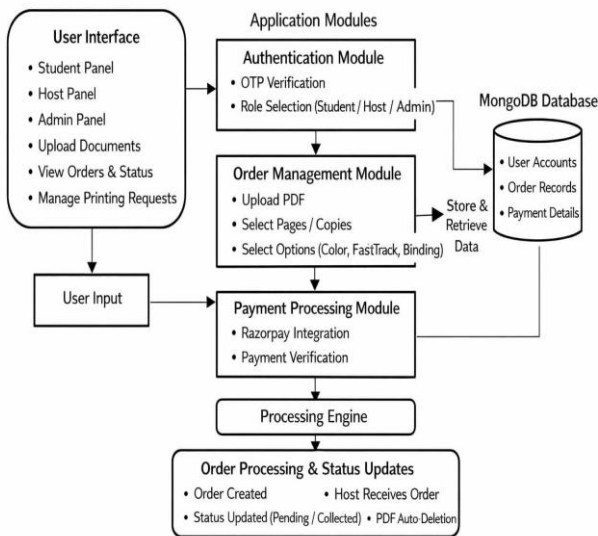


Figure 1: System Architecture

3.2 Order Processing and Priority Scheduling

It has a well-designed and efficient system of order processing to facilitate smooth management of the workflow. Once the student uploads a document and successfully executes the payment verification process, a unique Order ID is generated and the order is securely saved in the database. In the case of standard requests, the system uses the FIFO (First-In-First-Out) method to ensure fair scheduling. On the other hand, FastTrack orders have priority and are placed ahead of standard orders in the system. This ensures efficient scheduling and execution of the system's operations.

3.3 Payment Integration and Verification

For the system to perform online transactions securely, the Razorpay API has been incorporated to facilitate the same. Once the student initiates the payment process, the server creates a payment order and initiates the transaction using the API. Before the order is placed, the system verifies the transaction to ensure it has been successfully executed. No entry of the order is made to the database unless the payment transaction has been successfully verified. This ensures that only valid and successful transactions are recorded and saved securely in the system under a Payment ID. In addition to this, the revenue is also updated in the Admin dashboard, thus removing the need to rely on the traditional system of using cash.

3.4 Service Classification and Order Prioritization

The system categorizes the requests to be sent to the printer by classifying the type of service and the level of urgency required for efficient management of the workflow. Normal Service: For standard printing requests, the system uses the FIFO (First-In-First-Out) method to schedule the requests and process the print requests in the order they were received. FastTrack Service: Requests marked as FastTrack have priority over other requests and are sent to the printer before the normal requests, despite their receipt time. Premium Users: Premium users of the system can be provided with special services such as discounted pricing and quicker confirmation of their requests

3.5 Document Lifecycle and Privacy Management

For the sake of user privacy, the system has an automated document lifecycle management system where the PDF files uploaded by the users remain only on the server for a short period and then get deleted. Once the Host marks the order as "Collected," the system deletes the documents uploaded by the users after a specified period of time (e.g., 30 seconds).

3.6 Admin Analytics and Monitoring

The Admin module allows the monitoring of the entire system performance and business activities from a central point. The module provides valuable information such as the total number of orders, total revenue generated, FastTrack order statistics, premium user count, and daily, weekly, and monthly analytics. The system also uses data visualization tools such as Chart.js to provide graphical representations of revenue trends and order distribution, helping administrators analyze the system performance metrics effectively.

3.7 Authentication and Security Mechanism

The system uses an OTP-based security mechanism to authenticate the identity of the user during the login process. A one-time password is generated and sent to the registered email id or phone number, helping the user securely access the system. The system also uses role-based access control to restrict system functionalities according to the role assigned to the user. The student role is restricted to accessing only their own orders, the host role is restricted to updating the assigned orders, and the admin role is granted access to all functionalities and analytics systems.

3.8 Deployment and Web Interface

The proposed system is planned to be deployed as a web-based application that can be easily accessed by users without having to install any other software. The proposed system is planned to have an interactive interface that allows users to fill out forms, select options from dropdown menus, upload files securely, and display dynamic interfaces for Students, Hosts, and Admin users. The proposed system is planned to be highly scalable to accommodate future requirements of allowing hosts to register from multiple locations and implementing revenue management.

4. RESULT AND DISCUSSION

4.1 System Performance

The Smart On-Demand Xerox Management System was developed and tested in a live web environment, and it performed admirably in real-world usage. The system provided a seamless experience even after multiple uses, where the student could upload documents, securely pay, and track the order without experiencing delays. The API call to create an order, verify the payment, and refresh the dashboard consistently took less than two seconds, providing a seamless experience. The OTP login mechanism worked effectively without any unauthorized access during the test phase.

The integration with the payment gateway using Razorpay to verify the payment also ensured that the order would only be created after the payment had been made, effectively eliminating the revenue gap. The Host Dashboard also benefited from the FIFO and FastTrack features, which improved the efficiency with which the orders were being processed. The automated file deletion mechanism also helped delete the documents after they were uploaded 30 seconds after collection, improving the level of privacy. The system also improved the efficiency and transparency in the operations, including the document management, over the traditional manual Xerox process.

4.2 Discussion

The Smart On-Demand Xerox Management System demonstrates its true strength in streamlining the flow of work and the efficiency of services. Compared to the conventional manual print system, which involves creating orders, verifying payments, and scheduling tasks, Smart On-Demand Xerox Management System's automation significantly reduces downtime. The use of FIFO scheduling with FastTrack's priority scheduling ensures efficient service delivery. The tests also show that services can be delivered quicker, with precise tracking of revenues, and efficient coordination of services among the Student, Host, and Admin modules.

The secure payment verification with OTP-based login ensures that no unauthorized person can access the services or complete transactions. The automated deletion of documents also ensures the security of documents, which is a major concern in the management of digital documents.

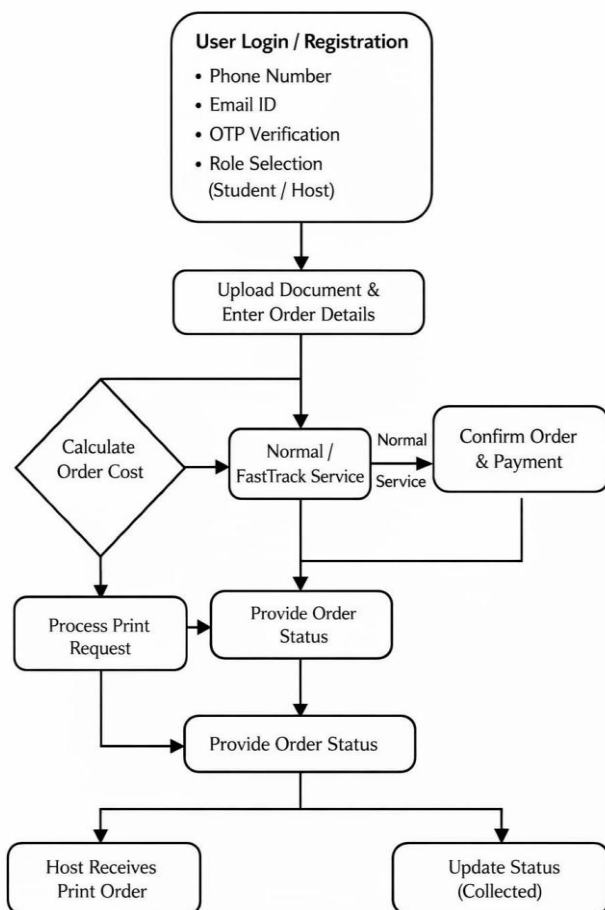


Figure:2 Flow Chart

Metric	Performance	Description
System Architecture	Client-Server Model	Web-based architecture using Node.js, Express.js, and MongoDB for scalable operation.
Authentication Success Rate	100%(Test Cases)	OTP-based login successfully validated users without unauthorized access
Payment Verification Accuracy	100%	Orders were generated only after successful Razorpay transaction verification.
Average API Response Time	< 2 seconds	Real-time order placement, payment confirmation, and dashboard updates.
Order Scheduling Method	FIFO + Priority Scheduling	Normal orders follow FIFO; FastTrack orders are prioritized automatically.
File Deletion Time	30 seconds after collection	Automated PDF deletion ensures document privacy and security.
Data Consistency	No Data Loss Observed	MongoDB successfully maintained persistent and accurate order records.
Admin Revenue Accuracy	100% Accurate Reporting	Revenue and order statistics dynamically updated in Admin dashboard.
Concurrent User Handling	Stable under multiple sessions	System handled simultaneous student and host logins without crashes.
Security Mechanism	Role-Based Access Control	Students, Hosts, and Admin have restricted and controlled access.

The real-time analytics dashboard for the Admin ensures that decisions can be taken based on the insights obtained on revenues, distribution of services, and usage. Although the current system is efficient for a single location, the planned upgrades to include features such as multi-location host registration, geographic service mapping, etc., can increase the scope of the system.



Figure: 3 Dashboard

Conclusion

The Smart On-Demand Xerox Management System presents a clear example of the capabilities of traditional printing services to be modernized with digital automation and web tech security features. The system's capabilities to upload documents remotely, verify payments online, and track orders in real-time facilitate efficiency and transparency in the services being offered. The system's ability to organize orders using FIFO and FastTrack priorities, and the automatic deletion of documents, also provide efficiency and security to the services.

The experiment results show improved efficiency in the operation, tracking of revenues, and communication between the Student, Host, and Admin components. The system also provides an opportunity to expand the system to various locations, commission-based revenues, and cloud-based systems. In conclusion, the system provides an efficient and effective platform to modernize the document printing services in educational institutions and commercial environments.

References

- [1] M. Tilkov and S. Vinoski, "Node.js: Using JavaScript to build high-performance network programs," *IEEE Internet Computing*, vol. 14, no. 6, pp. 80–83, Nov.–Dec. 2010, doi: 10.1109/MIC.2010.145.
→ Supports asynchronous backend processing used in your order handling system.
- [2] R. T. Fielding and R. N. Taylor, "Principled design of the modern Web architecture," *ACM Trans. Internet Technol.*, vol. 2, no. 2, pp. 115–150, May 2002, doi: 10.1145/514183.514185.
→ Foundation for REST APIs used between frontend and backend.
- [3] E. Brewer, "CAP twelve years later: How the 'rules' have changed," *Computer*, vol. 45, no. 2, pp. 23–29, Feb. 2012, doi: 10.1109/MC.2012.37.
→ Supports distributed data consistency concepts in MongoDB usage.
- [4] R. Elmasri and S. B. Navathe, *Fundamentals of Database Systems*, 7th ed. Boston, MA, USA: Pearson, 2016.
→ Database schema design and transaction handling.

- [5] H. Garcia-Molina, J. D. Ullman, and J. Widom, *Database Systems: The Complete Book*, 2nd ed. Upper Saddle River, NJ, USA: Pearson, 2009.

→ Data indexing and persistence models relevant to order storage.

- [6] L. Kleinrock, *Queueing Systems, Volume I: Theory*. New York, NY, USA: Wiley, 1975.

→ Theoretical basis for FIFO and priority scheduling in order processing.

- [7] R. Sandhu et al., "Role-based access control models," *IEEE Computer*, vol. 29, no. 2, pp. 38–47, Feb. 1996, doi: 10.1109/2.485845.

→ Supports role-based access (Student, Host, Admin modules).

- [8] OWASP Foundation, "Authentication cheat sheet," 2023. [Online]. Available: <https://owasp.org>

→ Security standards followed for OTP-based authentication.

- [9] M. Bishop, *Computer Security: Art and Science*. Boston, MA, USA: Addison-Wesley, 2018.

→ Secure file handling and privacy protection implementation.

- [10] P. Mell and T. Grance, "The NIST definition of cloud computing," *NIST SP 800-145*, 2011, doi: 10.6028/NIST.SP.800-145.

→ Supports scalability and future cloud deployment discussion.

- [11] A. Greenberg et al., "The cost of a cloud: Research problems in data center networks," *ACM SIGCOMM Comput. Commun. Rev.*, vol. 39, no. 1, pp. 68–73, Jan. 2009, doi: 10.1145/1496091.1496103.

→ Infrastructure scalability and web service deployment.

- [12] J. Nielsen, *Usability Engineering*. San Diego, CA, USA: Academic Press, 1993.

→ UI/UX principles used in dashboard and web interface design.

- [13] D. Hardt, "The OAuth 2.0 authorization framework," *IETF RFC 6749*, Oct. 2012.

→ Secure authorization concepts applied in session handling.

- [14] S. Nakamoto, "Bitcoin: A peer-to-peer electronic cash system," 2008.

→ Foundational concept of digital transaction verification relevant to payment gateways.

- [15] A. Silberschatz, P. B. Galvin, and G. Gagne, *Operating System Concepts*, 9th ed. Hoboken, NJ, USA: Wiley, 2013.

→ Process scheduling and resource management concepts applied in backend request handling.